Game design document

Island lesson collection

39 lessons / Nine distinct minigames / Three sequential worlds

Current reading edition | 11 November 2026

Documentation review. Unity development requires final owner approval.

This is the working game design for the owner's November 11, 2026 direction. It replaces the economy-led progression direction and incorporates the owner's subsequent confirmations: prerequisite order, three distinct minigames per world, sequential world unlocking, a required quiz, and offline AI teaching simulations. Detailed rules identified as implementation defaults are design recommendations chosen to make the specification executable; they are not additional owner confirmations. It describes the complete intended game, not a claim that the game is implemented or can be delivered within the remaining calendar time. **Documentation and preparation only: Unity development waits for the owner's explicit final approval.** Start here · Decisions · Lesson source pack

Canonical GDD | PDF editions

Generated from docs/game-design-lesson-collection.md. The Markdown source governs future revisions.

Pre-final: November 21, 2026. This design does not establish implementation or delivery capacity.

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Reading this edition

The core design appears in sections 1-13. Sections 14-15 contain every draft lesson panel and required question. Wide source tables are set as individual entries for legibility; the player journey is set as explicit transitions. Confirmed decisions and proposed implementation defaults retain their source labels.

1. The game in one paragraph

Chainy is an offline, single-player Android exploration game for ages 12-17, initially in English. The player selects an unlocked world, starting with Web3, explores a compact 3D island, and follows 13 ordered lesson quests. Each lesson leads to a collectible learning object or a distinct minigame encounter. Completing the lesson records its term in that world's encyclopedia collection and activates the next quest. Recording all 13 terms finishes its collection and opens a required multiple-choice review. Completing that review finishes the world and unlocks the next one: Web3 > Blockchain > AI.

2. Confirmed direction and implementation defaults

Status	Rule
Owner-directed	Main menu, world selector, then a 3D island with mobile joystick and buttons
Owner-directed	Speak to the tutorial NPC first to learn the island, controls, encyclopedia, quest log, and pause
Owner-directed	Explore freely after orientation; lesson completion follows one linear sequence within each world
Owner-directed	Exactly 13 lesson entries per world: 13 Web3, 13 Blockchain, and 13 AI, totaling 39
Owner-directed	Ordinary lessons are collectible objects; some lessons trigger unique minigames in separate scenes
Owner-directed	Minigames can require ordered item, NPC, and facility interactions
Owner-confirmed	Completing a lesson records its entry; after all 13 entries, a required quiz must be completed to finish the world
Prior owner direction retained	No functioning economy, shops, trading, or wallet balance system; these concepts can appear in fixed teaching examples
Owner-confirmed	Worlds unlock sequentially: Web3 first, then Blockchain, then AI
Owner-confirmed	Prerequisite-based lesson order and three distinct minigames per world, nine total
Owner-confirmed	Required review quiz; leaving before completion keeps the next world locked
Owner-confirmed	Offline AI teaching simulations are acceptable; no actual fitted model is required
Confirmed quiz rule	13 required questions per world, one per lesson; correct mistakes with feedback and retry until all 13 are resolved
Implementation default	Record first-response score separately from corrected completion; no percentage threshold, timer, or lives
Implementation default	Preserve successful checkpoints; unfinished minigames and abandoned quizzes restart their round on return

The existing source pack supplies concepts and draft assessment seeds. Its older five-clusters-per-world sequence, compulsory per-term quizzes, shared-purchase prerequisite, and AI-as-stretch framing do not govern this new

gameplay direction. Sections 14-15 provide concrete draft panel text and assessment content for implementation. Sean remains the documentation owner; these drafts have not received educator validation or established learning gains.

3. Player journey

From	Condition or action	To
Main menu	Continue	Select an unlocked world
Select an unlocked world	Continue	Load selected world and saved progress
Load selected world and saved progress	Continue	First visit: tutorial NPC
First visit: tutorial NPC	Continue	Explore island and follow active quest
Load selected world and saved progress	Returning visit	Explore island and follow active quest
Explore island and follow active quest	Continue	Ordinary lesson: inspect and record object
Explore island and follow active quest	Continue	Minigame lesson: ordered item, NPC, facility steps
Minigame lesson: ordered item, NPC, facility steps	Continue	Enter dedicated minigame scene
Enter dedicated minigame scene	Retry current round	Enter dedicated minigame scene
Enter dedicated minigame scene	Exit without completion	Explore island and follow active quest
Ordinary lesson: inspect and record object	Continue	Save lesson and encyclopedia entry
Enter dedicated minigame scene	Success	Save lesson and encyclopedia entry
Save lesson and encyclopedia entry	Save failed	Retry save; do not advance
Retry save; do not advance	Continue	Save lesson and encyclopedia entry
Save lesson and encyclopedia entry	Fewer than 13	Explore island and follow active quest
Save lesson and encyclopedia entry	13 recorded	Collection complete: quiz required
Collection complete: quiz required	Continue	Required 13-question review
Required 13-question review	Wrong answer: explanation and retry	Required 13-question review
Required 13-question review	Exit: next world remains locked	Select an unlocked world
Required 13-question review	All questions resolved and saved	World complete: unlock next world
World complete: unlock next world	Continue	Select an unlocked world

Each world supplies its own fictional examples and explanations. No lesson assumes that the player has bought an item or studied a term in a different world. World unlocking controls access; lesson examples remain self-contained and do not require a shared purchase or transferable inventory.

4. Screens and controls

Screen or state	What the player sees and can do
Main menu	Play opens world selection; Settings adjusts supported audio/accessibility settings; Quit exits on Android
World selection	Three named cards with recorded count, collection/quiz status, and Enter or Continue; locked cards explain which preceding world's quiz is required
Island exploration	Left joystick, contextual Interact button, current objective, Encyclopedia, Quest, and Pause buttons; fixed isometric camera
Tutorial/dialogue	Existing live-world dialogue overlay, short instructions and Next/Close choices; movement and world interactions blocked
Quest log	All 13 lessons in teaching order, completed entries, one current lesson, locked later lessons, and current step/location hint
Encyclopedia	World category tabs and 13 slots per category, recorded count, collected term explanations and examples
Lesson panel	Term name, short plain-language explanation, concrete example, and Record button
Minigame	Activity-specific touch controls, instructions, explanatory feedback, Retry, and Pause/Exit
Pause	Resume, controls reminder, sound settings, Return to world selection; no loss of already saved progress
Collection complete	13/13 recorded; Start required quiz, Keep exploring, or Return to world selection; next world stays locked
World completion	Collection and required quiz complete; Continue to next world or revisit this island; after AI show Journey complete
Review quiz	One multiple-choice question at a time; answer feedback, progress, exit, and final result
Loading/save failure	Clear status, Retry, and a safe route back where possible; no false completion or silent reset

Recommended encyclopedia discovery display: locked slots show their letter and “Not recorded”; explanatory content unlocks only after lesson completion. World categories can be browsed from any island, but browsing never grants completion. The quest order and encyclopedia's A-Z display order are separate: teaching can follow prerequisites while the collection remains alphabetical.

5. Tutorial and exploration rules

On the first visit to each world, the spawn area directs the player to its tutorial NPC. The player may move to the NPC, but lesson interactions stay locked until orientation finishes. The tutorial introduces movement, Interact, the active quest, collection, encyclopedia, and pause in short steps. Recommended interaction: ask the player to open and close Quest and Encyclopedia once, then finish orientation. Save tutorial completion separately for each world; offer the controls reminder on later visits. Closing before tutorial completion repeats orientation on return without granting lesson progress.

Retain the current visual direction: fixed orthographic isometric camera, restrained low-poly scenery, blue-haired yellow-clad Explorer, and screen-faced guides. Retain Sean as the Web3 guide, Matthew as the Blockchain guide, and Lee as the AI guide unless the owner changes them. Use a small island layout with visible landmarks and short routes, without jumping, combat, precision platforming, or a maze. Each current quest names its landmark and target object or NPC; a clearly distinguishable marker can guide exploration.

6. Lesson and collection rules

The following checkpoint details are proposed implementation rules for the confirmed loop.

Lesson type	Ordered interaction	Completion
Ordinary collection	Reach the active lesson object > Interact > inspect explanation/example > Record	Save the lesson completion and encyclopedia entry together
Minigame lesson	Follow its item/NPC/facility steps > launch its scene > satisfy its success rule > view explanation	Save minigame success, lesson completion, and encyclopedia entry together, then return to the island

Only the current lesson can advance. Interacting with a later lesson shows a hint to finish the current quest and leaves the object available. Earlier recorded objects may show their entry again; they never increment the count twice. Ordinary lessons do not insert an extra compulsory quiz or require touching the object accidentally. Record confirms collection, not that the learner read, understood, or mastered the explanation.

Minigame prerequisite objects are temporary quest evidence, not an inventory/equipment system. Store their completed step flags; show the acquired item in the quest log rather than adding inventory management. An out-of-order NPC or facility interaction explains the missing step without launching the scene. No required item can be sold, spent, dropped, or permanently lost.

Each successful lesson activates exactly the next lesson in the same world. The completion transaction updates the entry, lesson, next objective, and collection-complete flag consistently. It does not mark the world complete or unlock the next world before the required quiz is resolved and saved. Until saving succeeds, show an explicit pending/error state and retain a retry route. The object may visibly change to Recorded only when the checkpoint succeeds.

7. Complete lesson route

These tables preserve all 39 terms from the source pack in the prerequisite-based order selected for this implementation specification. They do not reduce the requirement to grouped lessons. The LC IDs are new identifiers; they do not reuse or reinterpret old V2 task or acceptance IDs. Once approved, keep each term's ID stable and store sequence separately if the order changes. Every entry uses the corresponding source-pack teaching point and page/source reference; props below are design proposals.

Web3 island: 13 lessons

ID	Source letter and term	Encounter	Type
LC-W01	F - Fiat Currency	Museum banknote display	Collect
LC-W02	C - Cryptocurrency	Fictional digital-coin placard	Collect
LC-W03	A - Altcoins	Coin comparison exhibit	Collect
LC-W04	X - XRP	Named-example display token	Collect
LC-W05	W - Wallets (Read-Only Awareness)	Address card > wallet guide > dummy terminal	Safe Address Handoff
LC-W06	I - Identity and Online Safety	Privacy shield card	Collect
LC-W07	Q - QR Codes in Web3	Non-scannable fictional poster	Collect
LC-W08	E - Exchanges (Know What, Not How)	Exchange information sign	Collect
LC-W09	J - Jupiter (JUP)	Named-service information plaque	Collect
LC-W10	V - Volatility (What It Means)	Chart strip > market guide > history display	Price Path
LC-W11	H - HODL	Slang postcard and explanation	Collect
LC-W12	R - Risks, Red Flags & Staying Scam-Safe	Envelope > attendant > message kiosk	Message Inspector
LC-W13	U - Understanding Web3 Safety	Guide's safety checklist	Collect

Blockchain island: 13 lessons

ID	Source letter and term	Encounter	Type
LC-B01	L - Ledger	Archive record book	Collect
LC-B02	B - Blockchain	Reference slip > archivist > workstation	Archive Reconstruction
LC-B03	D - Decentralization	Participant-network map	Collect
LC-B04	K - Keys (Public & Private)	Public/secret information display	Collect
LC-B05	O - On-Chain vs Off-Chain	Two-location record diagram	Collect
LC-B06	G - Gas Fees	Fixed example fee receipt	Collect
LC-B07	T - Tokens (Not Toys!)	Access-token information tile	Collect
LC-B08	N - NFTs (Non-Fungible Tokens)	Unique-token exhibit	Collect
LC-B09	S - Smart Contracts	Rule card > facility keeper > condition console	Condition Machine
LC-B10	M - Mining & Validators	Network-role exhibit	Collect
LC-B11	P - Proof Systems	Work/stake comparison board	Collect
LC-B12	Y - Your Data and Privacy in Web3	Public-record privacy panel	Collect
LC-B13	Z - Zero-Knowledge Proofs	Sealed credential > privacy guide > proof exhibit	Minimum Disclosure

AI island: 13 lessons

ID	Paired letter and AI term	Encounter	Type
LC-A01	A - Artificial intelligence	Helper information terminal	Collect
LC-A02	D - Dataset	Fictional example-card tray	Collect
LC-A03	F - Feature	Annotated example display	Collect
LC-A04	L - Label	Example tray > lab assistant > labeling desk	Label Lab
LC-A05	T - Training	Training-process illustration	Collect
LC-A06	M - Model	Learned-pattern diagram	Collect
LC-A07	C - Classification	Category-output panel	Collect
LC-A08	P - Prediction	Prediction-versus-result card	Collect
LC-A09	V - Validation	Fresh-case pack > lab guide > test station	Test the Helper
LC-A10	B - Bias	Missing-example display	Collect
LC-A11	H - Human oversight	Human-review terminal	Collect
LC-A12	G - Generative AI	Reference sheet > editor NPC > text exhibit	Fact Patch
LC-A13	R - Responsible AI	Responsible-use checklist	Collect

The AI terms are separately sourced additions, not chapters from the Web3 encyclopedia. Their matching letters do not make them equivalent to the original Web3/Blockchain terms. Named assets and services are informational examples, not endorsements, live tools, or investment instructions.

8. Nine distinct minigames

The owner confirmed three distinct minigames in each world. The nine scene assignments below are implementation defaults with different player rules; shared loading, pause, input, feedback, and save code is appropriate. There is no minigame selection screen that bypasses its lesson encounter.

Safe Address Handoff: LC-W05

After LC-W01-04, collect the fictional address card, speak to the wallet guide, then use the dummy terminal. The separate MG_SafeAddressHandoff scene is a routing puzzle, with a visible instruction that only the receiving address may leave the device. Three cards show a fictional receiving address, private signing key, and recovery phrase; all strings are visibly dummy examples and never resemble valid credentials. Tap the address card, then the Outgoing message slot; tap each secret card, then the Private storage slot. Submit the arrangement, then confirm the outgoing preview contains only the address. Success requires all three placements and the safe preview confirmation. An incorrect arrangement highlights which information would be exposed, keeps the puzzle open, and lets the player move that card. No network request, scanning, balance, payment, recovery flow, or actual wallet is present. After successful saving, record W and return beside the terminal with LC-W06 active.

Price Path: LC-W10

After LC-W01-09, collect a chart strip, speak to the market guide, then use the history display. The separate MG_PricePath scene is a graph-reading activity using fictional units and no trading choices. The player taps two time markers to select the largest adjacent change on each of two labeled charts: chart A has values 10, 12, 7, 8; chart B has values 6, 6, 9, 8. Chart A's correct interval is time 2-3 with a change of 5; chart B's is time 2-3 with a change of 3. Use displayed numerical differences rather than requiring rapid mental arithmetic. The final future marker is blank; choose “Unknown: past changes do not guarantee the next value” instead of an asserted rise or fall. Success requires both intervals and the uncertainty statement; mistakes display the selected values and permit reselection. After successful saving, record V and return beside the display with LC-W11 active.

Message Inspector: LC-W12

After LC-W01-11, collect a fictional envelope, speak to the kiosk attendant, then interact with the message kiosk. The separate MG_MessageInspector scene presents four fictional requests. For each, choose Continue, Verify, or Reject, then select the evidence supporting that choice.

Request	Expected action	Evidence
Known island guide asks the player to read a public island notice	Continue	Known source, public information, no secret or payment requested
Someone requests the player's secret to unlock free coins	Reject	Request for a secret
An unknown poster promises a prize through a link	Verify	Source/destination is unverified; no real link is opened
A message claims an asset's future value is guaranteed	Reject	Unsupported guarantee

Success requires the correct action and evidence for all four requests. An incorrect choice explains the overlooked evidence and allows retrying that request. The lesson teaches judgment about the displayed information; Continue

is not a universal guarantee that a real message is safe. After successful saving, record the R entry and return the player beside the kiosk with LC-W13 active.

Archive Reconstruction: LC-B02

After LC-B01, collect a reference slip, speak to the archivist, then use the archive workstation. The separate MG_ArchiveReconstruction scene shows checkpoint C0, three block cards, and a trusted reference for a fictional record field. The introductory panel explains a previous-block reference before asking the player to arrange the cards. Each card has a visible ID and previous-reference label: B1 follows C0, B2 follows B1, and B3 follows B2. Tap-select a card, tap its slot, and submit the order. Then compare two versions of a field against the trusted reference and select the altered field. Finally choose the explanation that a mismatch requires investigation against that reference.

Success requires the order, altered-field selection, and explanation to be correct. Feedback points to the specific mismatched reference or field and permits correction. This is a local record-linking and comparison exercise; labels are not cryptographic hashes, and a correct arrangement does not create consensus or prove historical truth. After successful saving, record the B entry and return beside the workstation with LC-B03 active.

Test the Helper: LC-A09

After LC-A01-08, collect a fresh-case pack, speak to the lab guide, then use the test station. The separate MG_TestHelper scene prominently says “Scripted AI teaching simulation.” It presents three unfamiliar fictional delivery cases, the verified case details, and authored helper predictions. For each case, reveal the prediction, decide whether it matches the verified case, and correct any wrong category.

Verified example	Authored prediction	Expected review
Complete address and matching destination; verified Routine	Routine	Matches
Destination mismatch; verified Review	Routine	Incorrect; change to Review
Missing address; verified Review	Review	Matches

The final choice asks why checking only familiar training examples is insufficient: new cases may expose mistakes. Success requires correct review of all three cases and that explanation. Feedback contrasts prediction with evidence and allows retry; no real training occurs in this candidate. These practice cases are not a validated evaluation set and the results do not measure a real model's reliability. After successful saving, record the V entry and return beside the test station with LC-A10 active. Actual model training is outside this confirmed simulation-based implementation.

Condition Machine: LC-B09

After LC-B01-08, collect a rule card, talk to the facility keeper, then use the condition console. The separate MG_ConditionMachine scene is a Boolean input-and-output puzzle, labeled “Contract-condition illustration; not a deployed smart contract.” Its displayed rule is “Open the demonstration locker only when permission is valid AND the request is unused.” For the three rounds, fixed inputs are valid/unused, valid/already used, and invalid/unused. Before running the illustration, select the expected Open or Stay closed lamp; the correct outputs are Open, Stay closed, Stay closed. After each selection, Run highlights both conditions and shows how the output follows the stated rule. Success requires all three correct predictions; an incorrect prediction receives the trace and retries that round. The example does not execute payment, mutate a wallet, or imply that all real contracts use this rule. After successful saving, record S and return beside the console with LC-B10 active.

Minimum Disclosure: LC-B13

After LC-B01-12, collect a sealed fictional credential, speak to the privacy guide, then use the proof exhibit. The separate MG_MinimumDisclosure scene is a disclosure puzzle, prominently labeled “Concept illustration; no cryptographic proof is generated.” Each round shows a verifier's question, a closed credential, an authored verification seal, and optional detail tiles. Round one asks only whether an invented resident meets an age threshold; choose the Meets threshold seal while leaving exact birthday and full name covered. Round two asks only whether an invented visitor has entry permission; choose the Permission confirmed seal while leaving the private credential number and home address covered. The seal is explicitly assumed to come from a suitable verification system; hiding text alone does not prove a claim. Submit succeeds only when the requested seal is selected and every unneeded detail remains covered, then choose “The verifier checks the requested claim without receiving the underlying details.” Unnecessary disclosure or a wrong seal gives specific feedback and resets only the current round. After successful saving, record Z, return beside the exhibit, and offer the required Blockchain quiz because all 13 entries are now recorded.

Label Lab: LC-A04

After LC-A01-03, collect the example tray, talk to the lab assistant, then use the labeling desk. The separate MG_LabelLab scene asks the player to attach known-answer stickers to four fictional training examples. Instructions show that a verified complete matching address is Routine, while a verified mismatch or missing address is Review. The four cases are complete/matching, missing address, complete/mismatched, and another complete/matching address; expected labels are Routine, Review, Review, Routine. Tap a sticker and then its example; submit the completed sheet. Incorrect stickers receive an evidence explanation and can be replaced individually. Success requires all four known labels and the final choice “A label is the known answer supplied with an example.” The scene states that this activity prepares labeled examples; it does not train a model or claim the player generated predictions. After successful saving, record L and return beside the desk with LC-A05 active.

Fact Patch: LC-A12

After LC-A01-11, collect the reference sheet, talk to the editor NPC, then use the text exhibit. The separate MG_FactPatch scene is a sentence-repair activity labeled “Authored example of an AI response; no live generation.” The reference sheet says “Museum opens at 09:00; entry is free; delivery desk closes at 16:00.” The authored response says “Museum opens at 08:00; entry is free; delivery desk closes at 16:00; visitors must send a password.” Tap the incorrect opening time and replace it with 09:00 from the reference; keep the two supported clauses; remove the unsupported password instruction. The final response must contain exactly the three supported clauses and no password request. Choose the explanation “Generated-looking text can contain errors or unsupported instructions; compare it with trusted information.” Success requires the corrected response and explanation; feedback identifies the unsupported clause and allows editing without restarting the scene. After successful saving, record G and return beside the exhibit with LC-A13 active.

Shared scene lifecycle

Save completed prerequisite steps before launch and remember the originating world, lesson, and safe return point. Disable repeated launch taps while loading; a load failure leaves the quest available to retry. Pause stops the activity; exit returns to the island without awarding that lesson. Completed prerequisite steps remain saved, so the player need not collect the same envelope/reference/case pack again. An unfinished minigame restarts its round on re-entry; mid-round resume is not promised. No timer, lives, spending, or permanent failure is required. A successful completion saves once before returning; if it fails, retain the result in memory and offer Retry Save or explicit discard of unsaved completion. If the app closes before a successful save, the lesson remains incomplete and the activity can be replayed. If the save succeeded but return/loading was interrupted, resume the island with the lesson already recorded. Revisiting a completed activity through its island encounter never creates a second entry or advances unrelated lessons.

9. Encyclopedia, completion, and review quiz

The encyclopedia is a collection record of 39 entries, divided into the three 13-entry categories. Each recorded entry has its letter, exact source term, short meaning, example, and source attribution. Use “Recorded” and “World complete” rather than “Mastered.” Unrecorded entries do not count toward completion; a title visible in the quest log is not a collected entry.

A world's collection completes when its 13 required lesson entries have been successfully saved. Offer its required quiz immediately after that save, including when the thirteenth lesson is a minigame. The player may postpone the quiz to explore or leave, but the world remains “Quiz required” and the next world stays locked. The world completes only when the quiz is resolved and that completion is successfully saved. Unlock Blockchain after Web3 completion and AI after Blockchain completion; new profiles start with only Web3 available. Completed worlds remain available for exploration, encyclopedia review, and replays. Finishing AI produces “Journey complete: 39 entries recorded and three world reviews completed,” without claiming mastery or measured learning improvement.

The owner confirmed 13 required questions per world, one for each lesson, with correction retries until all are correct. Section 15 defines every required question in lesson order. Four options, one correct answer, no timer, and no percentage threshold are the specified presentation rules. On the first submission for an item, record whether it was correct, then show the explanatory reason. For a wrong response, keep that item active, allow the learner to consult the encyclopedia, and ask them to choose again until it is correct. Only a correct choice enables Next; all 13 must be resolved before Finish can save review completion and unlock the next world. The result shows “First responses: X/13; review completed after corrections,” keeping initial performance distinct from assisted correction. If Finish saving fails, retain the answers and show Retry Save; the next world remains locked until the transaction succeeds.

Leaving early records an abandoned attempt separately and preserves encyclopedia progress; it grants no quiz completion or unlock. An abandoned quiz restarts all 13 questions on return; partial first responses remain part of that abandoned attempt, not a completed score. Replay creates a new attempt and preserves previous completion and unlocks even if replay is abandoned. No quiz result is represented as school grading, proof of learning gain, or a validated assessment score.

10. Persistence, recovery, and accessibility

Keep SQLite as the offline store and retain the simple local profile approach. No sign-in, teacher service, or network request is required to play. The minimal logical state is profile ID, content/schema version, per-world tutorial completion, active lesson/step, completed lesson IDs, encyclopedia entries, and separate quiz attempt/results including required-review completion. Collection completion derives from the 13 specified recorded lesson IDs, not raw interaction counts. World completion additionally requires a saved completed required review; access to the next world derives from that state. Save quiz completion and its unlock atomically so the world selector and persisted access cannot disagree. Minigame configuration and curriculum are authored content, not data supplied by a learner.

Store lesson completion and its encyclopedia entry atomically; duplicate taps and retries must be idempotent. On relaunch, use the last successful checkpoint and a safe island spawn near its objective, not an exact animation frame. Pause/encyclopedia/quest/dialogue overlays suspend world interaction and clear held joystick input before resuming. Android backgrounding pauses input and audio, attempts a safe checkpoint, and preserves the previous checkpoint if saving fails. Switching profiles clears the previous profile's in-memory progress before loading the next profile. Unknown or incompatible saved content must show a recovery message and preserve the old data rather than silently reset it. Existing saves are not automatically converted to 39-lesson completion; Christian must define and test any migration after reviewing the current schema.

Use readable text on the Android reference phone, large touch targets, text/icon feedback in addition to color, and equivalent essential information with sound muted. Use tap-to-select/place controls where dragging would require precision. The current baseline retains API 25 minimum installation and Android 10/API 29 reference testing; actual compatibility and performance require a named build and device checks.

11. Excluded systems

No functioning economy, shop, trading, real wallet, cryptocurrency transaction, seed entry, public blockchain, or live exchange is needed. Fixed examples may explain fees, wallets, tokens, or contract conditions without implementing those services. No general inventory/equipment system, combat, multiplayer, competitive leaderboard, paid hints, cloud chatbot, or standalone Wordle/BlockMatch mode is required by this design. Teacher collection, transfer, upload, and app updates remain separate from offline gameplay.

12. Design acceptance cases

These LC-G IDs define the new design's behavior; old V2-R and GP IDs keep their historical meaning.

ID	Observable check
LC-G01	Main menu opens the world selector; new profiles can enter only Web3, and locked cards explain the required preceding review
LC-G02	A new world's tutorial must finish before collecting lessons; completed orientation is remembered per world
LC-G03	Each world has exactly its 13 specified unique terms; all 26 original terms and 13 distinct AI terms are retained
LC-G04	Only the current lesson advances; later-object interaction cannot collect or remove it
LC-G05	Ordinary collection shows meaning/example and records exactly once after successful save

ID	Observable check
LC-G06	Each minigame launches only through its ordered lesson prerequisites and returns to the correct island
LC-G07	There are exactly nine separate minigame scenes, three per world, each with distinct rules, explanatory retry, and its stated success condition
LC-G08	Failed/cancelled/loading-interrupted minigames do not award completion; completed prerequisite steps persist
LC-G09	Save failure does not advance the quest or claim a recorded entry; repeated retry cannot duplicate progress
LC-G10	Thirteen entries produce Collection complete / Quiz required; only a saved resolved required review marks World complete and unlocks the next world
LC-G11	Wrong answers retry the same item with feedback; first-response scores, assisted completion, and abandonment remain distinct; a replay cannot revoke an existing unlock
LC-G12	Closing after a saved lesson and reopening offline restores the entry, next objective, and world count
LC-G13	Two synthetic local profiles remain separate across world changes, app relaunch, and profile switching
LC-G14	Joystick/world interaction is blocked beneath overlays and while paused; background/resume does not leave stuck input
LC-G15	UI remains usable on the named reference phone with audio muted and without color-only instructions
LC-G16	No real secrets, financial transactions, public blockchain, or cloud model are requested; AI simulation is labeled accurately
LC-G17	Completing the first two worlds unlocks Blockchain then AI, while all previously completed worlds remain accessible
LC-G18	A minigame Retry stays inside its scene; Exit returns to its originating island without completion and with prerequisite flags preserved
LC-G19	Failure to save review completion leaves the next world locked and preserves the completed answers for Retry Save

These checks require actual implementation evidence later and do not establish present completion.

13. Specification and content readiness

The owner's decisions about world order, required quizzes, prerequisite-based lessons, nine minigames, and scripted AI are settled. They are not remaining clarification gates. The 13-question correction policy is confirmed. The detailed prop assignments, locked-entry display, checkpoint rules, and exact activity data remain specified proposals for documentation review. They may guide implementation only after the owner's final approval to begin development. If those defaults change, update this specification and its acceptance cases together rather than silently changing the runtime.

The route, activity rules, draft 39 lesson panels, and draft 39-question bank are concrete in this document. All 13 items per world in section 15 define the mandatory review payloads; no quiz-writing decision is left to the runtime. Final educator review, reading-load testing, visual assets, Unity scene wiring, and named-device acceptance are not established by authored content. Do not claim that a spawned agent, a finished document, or a completed card supplies that evidence. Sean remains the named documentation owner while agents prepare the implementation artifacts.

Current build evidence and the November 10 session outcome inform preparation and evidence records. The owner has explicitly limited current work to documentation and necessary setup. Unity development and implementation-agent execution require the owner's final approval after reviewing this documentation. The execution plan governs the parallel work and integration checks. The November 21, 2026 pre-final remains visible; preserve integration, QA, manuscript, and rehearsal time without promising unmeasured completion.

14. Draft lesson panel payloads

Use the exact letter and title from section 7, the meaning and example below, and the corresponding citation/page locator in the lesson source pack. Ordinary encounters show these before Record; minigame encounters show them before launch and make them available in the activity's help panel. A minigame entry is recorded only after its success rule and completion save. All examples are fictional; named service/asset examples retain the source book's historical context and are not instructions to use or buy anything. The underlying encyclopedia PDF has not been checked for this draft; these paraphrases use the repository's contents inventory and source pack.

Web3 panel copy

LC-W01

Meaning: Fiat currency is money issued by a government; it can be physical cash or an ordinary bank balance.

Example: Philippine pesos in a bank account are still fiat even when shown on a screen.

LC-W02

Meaning: Cryptocurrency is a digital asset used on a cryptographic network; an in-game symbol is not automatically cryptocurrency or real money.

Example: The island's demonstration coin icon has no cash value and cannot be spent outside this lesson.

LC-W03

Meaning: Altcoins are cryptocurrencies other than Bitcoin; their names do not establish safety or future value.

Example: Two named coins can have different purposes without either being guaranteed to rise in price.

LC-W04

Meaning: XRP is a named cryptocurrency example in the source book, not government-issued money.

Example: Recognizing the name does not require opening an account or buying it.

LC-W05

Meaning: A wallet helps a user manage access and transactions; secret signing and recovery information must stay private.

Example: A fictional receiving address may be shared, while a private key or recovery phrase must not be sent in a message.

LC-W06

Meaning: Online safety includes protecting private information and checking who is making a request.

Example: A stranger saying “verification” does not make it safe to reveal your password.

LC-W07

Meaning: A QR code can encode a destination, but its appearance does not verify who controls that destination.

Example: A prize poster's code needs its source checked before anyone follows it; this game opens no real code.

LC-W08

Meaning: An exchange is a service where people trade digital assets; this lesson teaches recognition, not trading.

Example: “Swap one coin for another” describes an exchange service, not a wallet backup.

LC-W09

Meaning: The source book uses Jupiter as a named Solana swap-service example; it is not required to play Chainy.

Example: An informational plaque describes the example without connecting to the service.

LC-W10

Meaning: Volatility means a value can change substantially; a past increase does not guarantee a future one.

Example: A fictional chart rises from 10 to 12 and then falls to 7; its next point is unknown.

LC-W11

Meaning: HODL is slang for holding cryptocurrency, not a rule that guarantees a profit.

Example: A slogan saying “HODL means prices must rise” supplies no evidence about tomorrow's price.

LC-W12

Meaning: Requests for secrets, extreme urgency, suspicious links, and guaranteed rewards are warning signs.

Example: A message demanding a recovery phrase within one minute to claim free coins should be rejected.

LC-W13

Meaning: Pause, verify independently, and ask a trusted person when a Web3 request cannot be checked.

Example: If a new wallet link cannot be verified, do not proceed just because a stranger insists.

Blockchain panel copy

LC-B01

Meaning: A ledger is a record of activity; compare conflicting entries with an accepted reference before relying on them.

Example: A signed fictional reference says crate count 8 while one display says 6, so the discrepancy needs checking.

LC-B02

Meaning: A blockchain groups records into linked blocks under network rules; checking a link is different from proving that an event happened.

Example: The activity links B1 to checkpoint C0, B2 to B1, and B3 to B2, then checks a field against its reference.

LC-B03

Meaning: Decentralization distributes control or recordkeeping among participants instead of one operator.

Example: Three screenshots stored on one phone are still controlled on that phone; they are not three independent participants.

LC-B04

Meaning: Public receiving information can be shared; private signing information authorizes actions and must be protected.

Example: A demonstration address identifies where to receive, while its private key must remain secret.

LC-B05

Meaning: On-chain activity is recorded on a blockchain; off-chain activity happens elsewhere.

Example: Writing a private note in Chainy's local encyclopedia does not write anything to a public blockchain.

LC-B06

Meaning: Some blockchain actions require a network-use fee separate from the price of an item.

Example: A fixed example lists item price 8 and fee 1 as separate values; the game performs no purchase.

LC-B07

Meaning: A token can represent a digital unit, access right, or item; its meaning depends on the system.

Example: A fictional entry token may represent access without being real-world cash.

LC-B08

Meaning: An NFT is a unique token record; that record alone does not guarantee value, copyright, or ownership of an underlying file.

Example: A unique exhibit token does not guarantee that someone will buy it or that its holder owns the picture's copyright.

LC-B09

Meaning: A smart contract is deployed code that follows its programmed conditions when executed with relevant inputs.

Example: The lesson's locker opens only if permission is valid and the request is unused; the illustration is not a deployed contract.

LC-B10

Meaning: Networks can use different participants and methods to propose and check blocks.

Example: A network does not necessarily use both miners and validators doing the same job.

LC-B11

Meaning: Proof of Work and Proof of Stake are different approaches used in network agreement systems.

Example: A classroom tap or vote is not a real execution of either approach.

LC-B12

Meaning: Public addresses and records can expose activity; a pseudonym is not a guarantee of anonymity.

Example: An address with no printed name may still have publicly visible transactions.

LC-B13

Meaning: A suitable zero-knowledge protocol can establish a claim without disclosing its underlying private details.

Example: A verifier could learn that a threshold is met without receiving an exact birthday; this exhibit illustrates the idea without generating a proof.

AI panel copy

LC-A01

Meaning: AI systems can infer outputs such as predictions, recommendations, or generated content from inputs.

Example: A helper predicting a request category illustrates an AI task; Chainy's authored demonstration is labeled as a simulation.

LC-A02

Meaning: A dataset is a collection of examples; its quality and range matter.

Example: Three copies of one delivery request do not provide the same variety as three different relevant requests.

LC-A03

Meaning: A feature is an input detail a model can use to make a prediction.

Example: An address mismatch is relevant to reviewing a delivery request; the card's decorative color is not relevant in this example.

LC-A04

Meaning: A label is a known target answer supplied with a training example.

Example: A verified past request marked Review has Review as its label.

LC-A05

Meaning: During training, a learning method fits a model using examples.

Example: Correcting an example's label can affect fitting; this scripted exhibit explains that process but does not train a model.

LC-A06

Meaning: A trained model uses learned patterns to infer an output for a new input.

Example: A fixed list of authored replies that never depends on examples is not a model trained by the player.

LC-A07

Meaning: Classification predicts which defined category an input belongs to.

Example: Predicting Routine or Review for a request is classification, not image generation.

LC-A08

Meaning: A prediction is a proposed output that can be wrong and should be checked against appropriate evidence.

Example: A helper's Routine suggestion disagrees with a verified Review case, so the suggestion needs correction.

LC-A09

Meaning: Validation checks performance on examples separate from training; final test examples should stay untouched until final evaluation.

Example: Getting familiar cards right does not establish that the helper handles new requests correctly.

LC-A10

Meaning: Missing or skewed examples can contribute to systematically uneven results.

Example: Adding more daytime requests may not fix errors on night requests when night examples are missing.

LC-A11

Meaning: Human oversight means people retain responsibility and check consequential suggestions.

Example: A person pauses a suggested dispatch when the address is missing, even if the helper says Routine.

LC-A12

Meaning: Generative AI can produce text, images, and other content, but plausible content may contain errors.

Example: An authored response says a museum opens at 08:00 while its trusted notice says 09:00; check the notice.

LC-A13

Meaning: Responsible AI considers purpose, privacy, possible harm, fairness, and human control.

Example: Practice with fictional examples instead of collecting real learner names when they are unnecessary.

15. Draft assessment bank and required review payloads

Question IDs are Q- followed by the corresponding lesson ID. The tables give exact draft wording, four ordered options, the correct option, and feedback. They paraphrase the corresponding source-pack checks and are implementation content, not a validated instrument. Options do not depend on live prices, actual accounts, personal information, or external websites.

The mandatory 13-question payloads follow the corresponding lesson order:

- Web3: Q-LC-W01 through Q-LC-W13.
- Blockchain: Q-LC-B01 through Q-LC-B13.
- AI: Q-LC-A01 through Q-LC-A13.

Every question below is required in its world's review. A new attempt starts at question 1, including after abandoning an earlier attempt. Wrong answers are corrected within that attempt; they cannot be skipped to unlock a world.

Web3 question bank

Q-LC-W01

Which example is fiat currency?

A: A fictional game badge;

B: Philippine pesos in a bank account;

C: A private key;

D: An NFT picture

Correct and feedback: B. Government-issued money remains fiat when displayed digitally.

Q-LC-W02

The island displays a fictional coin teaching prop. What does that establish?

A: It is real cryptocurrency;

B: It can be spent at a shop outside the game;

C: It is a teaching symbol with no promised cash value;

D: Its value will rise

Correct and feedback: C. The example does not create a real financial asset.

Q-LC-W03

What does “altcoin” refer to in this lesson?

A: A cryptocurrency other than Bitcoin;

B: Any government banknote;

C: A password backup;

D: A guaranteed investment

Correct and feedback: A. The name identifies a broad category, not safety or value.

Q-LC-W04

How is XRP used in this course?

A: A required purchase;

B: Government-issued money;

C: A way to unlock the island;

D: A named cryptocurrency example

Correct and feedback: D. Recognition is sufficient; no purchase or endorsement is involved.

Q-LC-W05

Which item may be included in the fictional receiving message?

A: The private signing key;

B: The receiving address;

C: The recovery phrase;

D: The account password

Correct and feedback: B. Receiving information can be shared; signing and recovery secrets must stay private.

Q-LC-W06

A stranger asks for a password “to verify you.” What is the safe response?

A: Refuse and check through a trusted route;

B: Send half the password;

C: Send it if they hurry;

D: Post it publicly

Correct and feedback: A. A verification claim does not justify sharing a secret.

Q-LC-W07

An unknown poster offers a prize through a QR code. What should happen first?

A: Follow it immediately;

B: Enter a secret to see the prize;

C: Share it with everyone;

D: Check the source and destination through a trusted route

Correct and feedback: D. The code's appearance does not establish trust.

Q-LC-W08

A service swaps one digital asset for another. What is it describing?

A: A private key;

B: A school attendance list;

C: An exchange service;

D: A wallet backup phrase

Correct and feedback: C. Trading and credential recovery are different functions.

Q-LC-W09

Must a learner use Jupiter to complete Chainy?

A: Yes, to pay the guide;

B: No, it is an informational example;

C: Yes, to save progress;

D: Yes, to take the quiz

Correct and feedback: B. The source example does not require connecting to a service.

Q-LC-W10

A fictional asset rose in value yesterday. What follows about tomorrow?

A: It must rise again;

B: It cannot fall;

C: Its next value remains uncertain;

D: Its value is guaranteed by its name

Correct and feedback: C. Past movement does not guarantee a future price.

Q-LC-W11

What does the slogan HODL establish?

A: It is slang for holding, without a guaranteed outcome;

B: A legal promise of profit;

C: A private key;

D: A fixed future price

Correct and feedback: A. Slang is not evidence about future value.

Q-LC-W12

A message demands a recovery phrase within one minute for free coins. Which response is justified?

A: Send it before time runs out;

B: Send only some words;

C: Trust the free reward claim;

D: Reject the request because it asks for a secret

Correct and feedback: D. The secret request and urgency are warning signs.

Q-LC-W13

You cannot verify a new wallet link. What is the next safe action?

A: Proceed because it looks professional;

B: Enter a password to test it;

C: Pause and seek trusted help;

D: Assume every link is safe

Correct and feedback: C. Uncertainty calls for checking, not disclosure.

Blockchain question bank

Q-LC-B01

A reference says count 8, while a local record says 6. What should you do?

A: Automatically accept 6;

B: Investigate the discrepancy against the reference;

C: Delete all records;

D: Assume both describe the same value

Correct and feedback: B. A mismatch calls for checking before reliance.

Q-LC-B02

What does correctly arranging the activity's linked block cards show?

A: A real network reached consensus;

B: Every event in history is true;

C: A public blockchain was created;

D: The displayed previous-reference order is consistent

Correct and feedback: D. Local linking does not establish network agreement or historical truth.

Q-LC-B03

Are three screenshots on one phone three independent network participants?

A: No, they are copies controlled on one device;

B: Yes, because there are three pictures;

C: Yes, if colored differently;

D: Only if renamed

Correct and feedback: A. Multiple local copies do not create independent control.

Q-LC-B04

Which information must remain secret in the example?

A: A public notice;

B: The receiving address;

C: The private signing key;

D: The lesson title

Correct and feedback: C. Private signing information can authorize actions.

Q-LC-B05

A private encyclopedia note is stored only on this phone. Where is it recorded?

A: On every public blockchain;

B: Off-chain in the local app;

C: In a public consensus vote;

D: In a real wallet transaction

Correct and feedback: B. Local storage does not create an on-chain record.

Q-LC-B06

A fixed receipt shows price 8 and network fee 1. What is the fee?

A: Proof the item is free;

B: The item's name;

C: A secret recovery word;

D: A separate transaction cost

Correct and feedback: D. The displayed fee is separate from the listed item price.

Q-LC-B07

Must every token have real-world cash value?

A: Yes, if drawn as a coin;

B: Yes, if rare;

C: No, its meaning and rights depend on the system;

D: Yes, if collected

Correct and feedback: C. A token can represent access or another unit without cash value.

Q-LC-B08

What does a unique NFT record guarantee by itself?

A: None of resale value, copyright, or ownership of an underlying file;

B: A guaranteed profit;

C: Automatic copyright ownership;

D: Ownership of every copy of a picture

Correct and feedback: A. A token record alone does not establish those additional rights or outcomes.

Q-LC-B09

The locker rule requires valid permission AND an unused request. Permission is valid, but the request was used. What follows?

A: Open because one condition passed;

B: Open if the player taps twice;

C: Choose randomly;

D: Stay closed because both conditions are required

Correct and feedback: D. The displayed rule requires both conditions to be true.

Q-LC-B10

Must every blockchain use miners and validators doing the same job?

A: Yes, always;

B: No, roles depend on the network design;

C: Only on phones;

D: Only in a game

Correct and feedback: B. Networks use different methods and participant roles.

Q-LC-B11

Does tapping a card perform real Proof of Work or Proof of Stake?

A: Yes, every tap creates a block;

B: Yes, if the card glows;

C: No, it is only a teaching interaction;

D: Yes, after collecting thirteen cards

Correct and feedback: C. An illustration does not execute a public network's agreement system.

Q-LC-B12

A public address has no printed real name. Is its activity necessarily private?

A: No, activity can still be visible or linked;

B: Yes, a pseudonym hides everything;

C: Yes, all ledgers are private;

D: Only the owner can ever see it

Correct and feedback: A. A pseudonym is not a promise of anonymity.

Q-LC-B13

What could a suitable zero-knowledge system establish in the exhibit's example?

A: Everyone's birthday;

B: A guaranteed identity from hidden text alone;

C: A cryptocurrency price;

D: That a threshold is met without revealing the exact birthday

Correct and feedback: D. A suitable protocol checks the claim; this game only illustrates that idea.

AI question bank

Q-LC-A01

Which task illustrates prediction by an AI helper?

A: Always printing a fixed title;

B: Predicting a request's category from input details;

C: Closing a menu;

D: Drawing a static button

Correct and feedback: B. Prediction uses input information to propose an output; Chainy's example remains labeled as a simulation.

Q-LC-A02

What improves the range of a practice dataset?

A: Copying one card repeatedly;

B: Changing only its border color;

C: Including different relevant verified cases;

D: Removing every difficult example

Correct and feedback: C. Relevant variety adds coverage that duplicate copies do not.

Q-LC-A03

Which feature is relevant to reviewing the fictional delivery request?

A: Whether its destination mismatches;

B: Decorative card color;

C: The screen border;

D: The player's favorite music

Correct and feedback: A. Destination mismatch bears on this task's decision.

Q-LC-A04

A verified past request is marked Review. What is Review here?

A: A destination address;

B: A private password;

C: A new prediction;

D: The example's known label

Correct and feedback: D. A label supplies the known target answer for an example.

Q-LC-A05

What happens during actual model training?

A: Every answer is guaranteed correct;

B: A learning method fits a model using examples;

C: A public blockchain is repaired;

D: An authored caption becomes a trained model

Correct and feedback: B. Fitting depends on examples; showing a scripted answer is different.

Q-LC-A06

The game repeats authored replies regardless of supplied examples. Has the player trained a model?

A: Yes, because a robot appears;

B: Yes, because a label was tapped;

C: No, the replies do not result from fitting to those examples;

D: Yes, if the scene is named AI

Correct and feedback: C. A simulation must not claim example-dependent training it does not perform.

Q-LC-A07

Predicting Routine or Review is which task?

A: Classification;

B: Image generation;

C: Mining;

D: Wallet recovery

Correct and feedback: A. Classification selects among defined categories.

Q-LC-A08

A helper predicts Routine, but the verified case is Review. Which should guide this check?

A: The brighter button;

B: The first prediction forever;

C: The robot's appearance;

D: The verified case evidence

Correct and feedback: D. A prediction can be wrong and must be checked.

Q-LC-A09

A helper gets every familiar training example right. Is that enough to claim it handles new cases?

A: Yes, familiar examples prove everything;

B: No, check separate unfamiliar cases and inspect mistakes;

C: Yes, if it is fast;

D: Yes, if the cards are duplicated

Correct and feedback: B. Performance on familiar examples alone does not establish generalization.

Q-LC-A10

All examples are daytime requests and the helper often fails at night. What should be examined?

A: Only the card color;

B: How to copy daytime cards faster;

C: Whether relevant night examples are missing;

D: How to hide the failures

Correct and feedback: C. Missing coverage can contribute to uneven results.

Q-LC-A11

A helper suggests dispatch despite a missing address. What should the person do?

A: Pause and check before allowing dispatch;

B: Send automatically;

C: Invent an address;

D: Treat the suggestion as proof

Correct and feedback: A. Consequential actions require human judgment and adequate information.

Q-LC-A12

An authored AI response says 08:00, while the trusted museum notice says 09:00. What should the editor do?

A: Trust the response because it sounds confident;

B: Delete the trusted notice;

C: Assume both times mean the same thing;

D: Correct the response using the notice

Correct and feedback: D. Plausible generated-looking text can conflict with reliable evidence.

Q-LC-A13

Fictional examples are enough for practice. Should the game collect real learner identities for that dataset?

A: Yes, more personal information is always better;

B: No, avoid unnecessary personal data;

C: Yes, to make the robot look realistic;

D: Yes, without telling anyone

Correct and feedback: B. Responsible design uses only the data needed for the task.